Could missingness, duplicate rows, or recording practices distort our conclusions?

The quality audit identified 33 missing values in total: 27 in height and 6 in weight. It also found 24 repeated rows beyond the first identical occurrence. No constant or entirely missing columns were detected. These issues were flagged for review, but duplicate rows were retained because identical recorded attributes do not necessarily represent erroneous records. Missing values and recording practices should be considered when interpreting the analysis.

Are the classes equally represented?

The classes are almost equally represented. The majority class accounts for 50.53% of rows, while the other class accounts for 49.47%. A classifier always predicting the majority class would achieve 50.53% accuracy, but this is only a baseline, not a validated model result.

where are values concentrated, how wide is the range, and are there skewness, gaps, spikes, or multiple peaks?

The numerical features show different distributions and ranges. Age is concentrated around middle and older adulthood, while height and weight show their own patterns and some missing values. Blood pressure values cover a range of recorded measurements. Histograms help identify concentrations, gaps, and possible spikes, but these patterns alone do not show differences between cardiovascular disease classes.

which categories are common or rare, and is an unknown category substantial?

The categorical features are unevenly distributed. Gender category 1 is more common than category 2. Cholesterol and glucose level 1 dominate their respective categories, while smoking and alcohol use are less common. Most records indicate physical activity (active = 1). These differences in category frequencies should be considered when comparing cardiovascular disease rates.

which measurements have unusual values relative to their distribution?

Boxplots identify possible outliers in all five numerical features. Height and weight contain unusually low and high values, while systolic (ap_hi) and diastolic (ap_lo) blood pressure also show values beyond the whiskers. A few low age values are flagged as well. These observations were retained because outliers are not necessarily errors; their plausibility and potential importance for classification should be assessed before deciding whether to remove them.

does each class occupy a different part of a measurement's range?

- **age_years**: The distributions show a clear shift. Younger age groups (e.g., under 48) are heavily dominated by healthy individuals (blue line), while cardiovascular disease cases (orange line) increase noticeably in older age brackets (55 to 65 years).
- **ap_hi & ap_lo**: Both blood pressure metrics display significant separation. Healthy individuals concentrate strongly around normal baseline values (ap_hi ~120, ap_lo ~80), whereas cardiovascular disease patients shift distinctly toward higher blood pressure ranges.
- **weight**: Shows a high degree of overlap between both classes, though cardiovascular disease cases exhibit a slight upward shift toward heavier weight ranges.
- **height**: Both classes share almost identical distributions, indicating that height alone does not occupy a distinct range or separate the target classes effectively.

Which measurements have unusual values relative to their distribution, and how do classes compare in terms of medians, spread, and outliers?

- **age_years**: The median age for the cardiovascular disease group is notably higher (~56 years) compared to the healthy group (~52 years), showing a clear upward shift with age.
- **ap_hi & ap_lo** : Both metrics exhibit substantial differences in medians and spreads. The cardiovascular disease class shows significantly higher median systolic (ap_hi) and diastolic (ap_lo) pressures with wider interquartile ranges, alongside numerous flagged outliers in both classes.

- **weight**: Displays a high degree of overlap in medians between the two classes, though the cardiovascular disease group exhibits a slight upward shift and a higher concentration of extreme high-end outliers.
- **height**: Both classes share nearly identical medians, spreads, and whiskers, indicating that height has minimal utility for separating the target classes.

How do the average and observed extremes differ between target categories?

- **age_years**: The mean age and observed minimum age are visibly higher in the cardiovascular disease group compared to the healthy group (~55 vs ~51 for mean; ~39 vs ~29 for min). However, both target classes share the exact same maximum age of ~65 years.
- **ap_hi & ap_lo** : Both mean systolic and diastolic blood pressures are higher in the cardiovascular disease group. The minimum recorded values are also higher for diseased patients in ap_hi, while maximum observed values reach high extremes (~240 for ap_hi and ~150 for ap_lo) in both classes.
- **weight**: The mean weight is slightly higher for the cardiovascular disease group (~76 kg vs ~71 kg), whereas both minimum (30 kg) and maximum (200 kg) observed values are identical across both categories.
- **height**: Shows virtually identical values across both classes for mean (~164 cm), minimum (100 cm), and maximum (~200 cm), confirming that height does not differentiate between target classes.

How many rows from each class occur in each category, and among rows in a category, what percentage belongs to each target class?

- **cholesterol & gluc**: Show strong predictive power; as cholesterol rises to level 3, cardiovascular disease prevalence increases sharply to nearly **76%**.
- **gender, smoke, & alco**: Rates remain virtually even (~50%) across categories, showing minimal individual discriminative power.
- **active**: Physical inactivity (active = 0) exhibits a slightly higher disease rate (~53%) compared to active individuals.